

c) $2+1=3$ Ans: $2+1 \neq 3$

d) The summer in Maine is hot and sunny.

Ans: It is not the case that the summer in Maine is hot and sunny.

6. What is the negation of each of these propositions?

a) Jennifer and Teja are friends.

Ans: Jennifer and Teja are not friends.

b) There are 13 items in a baker's dozen.

Ans: There are either more than or less than 13 items in a baker's dozen.

c) Abby sent more than 100 text messages yesterday.

Ans: Abby did not send more than 100 text messages yesterday.

d) 121 is a perfect square.

Ans: 121 is not a perfect square.

7. What is the negation of each of these propositions?

(a) Steve has more than 100 GB free disk space on his laptop.

Ans: Steve does not have more than 100 GB free disk space on his laptop.

(b) Zach blocks e-mails and texts from Jennifer.

Ans: It is not the case that Zach blocks emails and texts from Jennifer.

(c) $7 \cdot 11 \cdot 13 = 999$

Ans: $7 \cdot 11 \cdot 13 \neq 999$

(d) Diane rode her bicycle 100 miles on Sunday.

Ans: It is not the case that Diane rode her bicycle 100 miles on Sunday.

8. Suppose that Smartphone A has 256 MB RAM and 32 GB ROM, and the resolution of its camera is 8 MP; Smartphone B has 288 MB RAM and 64 GB ROM, and the resolution of its camera is 4 MP; and Smartphone C has 128 MB RAM and 32 GB ROM, and the resolution of its camera is 5 MP. Determine the truth value of each of these propositions.

(a) Smartphone B has the most RAM of these three smartphones $\rightarrow$ True

(b) Smartphone C has more ROM or a higher resolution camera than Smartphone B $\rightarrow$ True